SINGAPORE DIMENSIONS • GRADE 3A • CHAPTER 3

# Multiplication & Division: Arrays & Groups

Aligned to CCSS.MATH.3.OA.A.3 • Guided Concrete-Pictorial-Abstract (CPA) Investigation

Student Name: _____ Date: _____ Class: _____ **Score: ___ / 20**

**M** Primary Manipulative: **Interlocking Cubes & Gridfinity Baseplate**

**C** Required Calculator: **SC-6 Elementary (Grades K-6)**

**WORKED EXEMPLAR: A classroom has 7 rows of 4 desks. Total desks?**

**1. Concrete Model (Desk)**

Snap cubes into 4 rows of 6 columns on the Gridfinity dock.

**2. Pictorial Model (Paper)**

Sketch a rectangular area array labeled 4 and 6.

**3. Abstract Stack (LCD)**

Multiply rows by columns. On SC-6, press [6] [REP] to hear tactile click.

**Keystroke Flow:**



**1** **36 stickers shared equally by 4 students.**

Manipulative Sketch / Bar Model:

Calculator Keystroke Flow:

Prediction: _____

**Final Answer:** _____

**2** **9 bags with 5 apples each. Total apples?**

Manipulative Sketch / Bar Model:

Calculator Keystroke Flow:

Prediction: _____

**Final Answer:** _____

**3** **42 markers divided into boxes of 6.**

Manipulative Sketch / Bar Model:

Calculator Keystroke Flow:

Prediction: _____

**Final Answer:** _____

**4** **6 rows of 8 chairs. Total chairs?**

Manipulative Sketch / Bar Model:

Calculator Keystroke Flow:

Prediction: _____

**Final Answer:** _____

TEACHER FACILITATOR GUIDE • SINGAPORE DIMENSIONS • GRADE 3A

# Official Answer Key & Solutions: Chapter 3

Multiplication & Division: Arrays & Groups • Canonical RPN Keystroke Sequences & Diagnostic Guide

| # | Problem Prompt | Canonical Keystrokes | Expected Result |
|---|---|---|---|
| 1 | **36 stickers shared equally by 4 stude**<br>Standard: CCSS.MATH.3.OA.A.3 | [3][6] [ENTER] [4] [÷] | 9 stickers each |
| 2 | **9 bags with 5 apples each. Total appl**<br>Standard: CCSS.MATH.3.OA.A.3 | [9] [ENTER] [5] [×] | 45 |
| 3 | **42 markers divided into boxes of 6.**<br>Standard: CCSS.MATH.3.OA.A.3 | [4][2] [ENTER] [6] [÷] | 7 boxes |
| 4 | **6 rows of 8 chairs. Total chairs?**<br>Standard: CCSS.MATH.3.OA.A.3 | [6] [ENTER] [8] [×] | 48 |

## TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

**Key Misconception to Target:**

Division is the inverse: total in Y, group size in X, [÷] yields row count.

**Why RPN Eliminates This Math Barrier:**

Traditional in-fix calculators require nested parentheses and forward-planning. StackCalc's live 4-line LCD display holds operands in physical registers (Y and X), matching the tangible state of the Interlocking Cubes & Gridfinity Baseplate. Students see operations consume values immediately, eliminating syntax anxiety.

## CLASSROOM SCORING & MASTERY RUBRIC

**Exemplary (18–20 pts):** Correct manipulative model drawn; accurate RPN keystrokes; zero syntax errors; precise a

**Proficient (14–17 pts):** Correct numerical result; minor sketch irregularity or unneeded ENTER key pressed.

**Developing (10–13 pts):** Understands concrete model but misidentifies register Y vs X order (operand inversion).

**Intervention (<10 pts):** Needs additional hands-on tactile practice with physical manipulative blocks before calc